

Daily Geometry Drill #4 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Section A Rapid Recognition

1. Answer: "(3, 3)"

Method: The centroid is the average of the three vertices: $(\frac{0+6+3}{3}, \frac{0+0+9}{3}) = (3, 3)$.

Investigate further: Forgetting to divide every coordinate by three, or averaging only two vertices.

2. Answer: "(3, 4)"

Method: In a right triangle the circumcentre is the midpoint of the hypotenuse, the midpoint of (6, 0) and (0, 8) is (3, 4).

Investigate further: Choosing the midpoint of the wrong side, or halving only one coordinate.

3. Answer: 1

Method: The incircle is tangent to all three sides. For a right triangle with legs 3, 4 the hypotenuse is 5 and $r = \frac{a+b-c}{2} = \frac{3+4-5}{2} = 1$.

Investigate further: Writing $\frac{a+b+c}{2}$ (that is s , the semiperimeter) instead of $\frac{a+b-c}{2}$.

4. Answer: 7

Method: Legs 6, 8 give hypotenuse 10. The inradius is $\frac{6+8-10}{2} = 2$ and the circumradius of a right triangle is half the hypotenuse, $R = 5$, so $r + R = 7$.

Investigate further: Using $R = \frac{abc}{4A}$ with the wrong area, or adding r to the hypotenuse instead of R .

5. Answer: 5

Method: $A = \frac{1}{2} \cdot 6 \cdot 8 = 24$ and $R = \frac{abc}{4A} = \frac{6 \cdot 8 \cdot 10}{4 \cdot 24} = 5$. Equivalently the hypotenuse 10 is a diameter, so $R = 5$.

Investigate further: Treating the circumdiameter as a side; the hypotenuse is twice R , not equal to it.

6. Answer: "(3, 2)"

Method: $(\frac{1+5+3}{3}, \frac{2+6-2}{3}) = (3, 2)$.

Investigate further: Mixing up the x and y sums.

7. **Answer:** "(0, 0)"

Method: A right triangle has its orthocentre at the right-angle vertex, which is the origin here.

Investigate further: Substituting for the other two altitudes and getting tangled; the right angle decides it.

8. **Answer:** 6

Method: The angle bisector theorem: $BD : DC = AB : AC = 2 : 3$ with $BD + DC = 10$, so $DC = \frac{3}{2+3} \cdot 10 = 6$.

Investigate further: Using the ratio the wrong way round and giving 4. The smaller side segment pairs with the smaller adjacent side.

9. **Answer:** 4

Method: The median to the base is the altitude: $\sqrt{5^2 - 3^2} = \sqrt{16} = 4$.

Investigate further: Dropping only half, not the full base, or halving the base incorrectly.

10. **Answer:** 8

Method: The centroid is $\frac{2}{3}$ of the way from the vertex, so the vertex-to-centroid distance is $\frac{2}{3} \cdot 12 = 8$.

Investigate further: Using the $\frac{1}{3}$ tail (the centroid-to-midpoint piece) and answering 4.

Section B Manipulation Drills

1. **Answer:** C) (4, 3)

Method: $(\frac{2+4+6}{3}, \frac{1+3+5}{3}) = (4, 3)$.

Investigate further: Spotting equal spacings and guessing the middle point without summing.

2. **Answer:** B) 2

Method: $r = \frac{5+12-13}{2} = 2$. Sanity check via $A = rs$: area 30, semiperimeter 15, $r = 2$.

Investigate further: Using s in place of the triangle inequality expression.

3. **Answer:** B) $\frac{25}{2}$

Method: $25 = 7^2 + 24^2$, so this is a right triangle: hypotenuse 25 is the diameter, $R = \frac{25}{2}$.

Investigate further: Answering the hypotenuse itself, 25, instead of the radius $\frac{25}{2}$.

4. **Answer:** 9

Method: The centroid sits $\frac{2}{3}$ along the median: $AG = \frac{2}{3}AM$, and $AM = \frac{3}{2} \cdot 6 = 9$.

Investigate further: Reversing the fraction and getting 4.

5. **Answer:** B) (4, 3)

Method: Right angle at $(0,0)$, so the circumcentre is the midpoint of the hypotenuse $(8,0)$ – $(0,6)$, namely $(4,3)$.

Investigate further: Halving the perpendicular side midpoints and landing on a quadrant distractor.

6. **Answer:** B) 6

Method: By the angle bisector theorem $\frac{BD}{DC} = \frac{AB}{AC}$, so $DC = \frac{BD \cdot AC}{AB} = \frac{4 \cdot 9}{6} = 6$.

Investigate further: Multiplying BD by the *near* side AB instead of the *far* side AC .

7. **Answer:** B) 8

Method: The apex median is the altitude: $\sqrt{10^2 - 6^2} = \sqrt{64} = 8$.

Investigate further: Substituting the wrong leg of the right triangle (6 twice instead of once).

8. **Answer:** 4

Method: $r = \frac{A}{s} = \frac{84}{21} = 4$.

Investigate further: Using the full perimeter 42 as the divisor and getting 2.

9. **Answer:** B) 3

Method: Hypotenuse is $\sqrt{8^2 + 15^2} = 17$; $r = \frac{8+15-17}{2} = 3$.

Investigate further: Answering $\frac{17}{2}$ as if the question asked for R , not r .

10. **Answer:** $\frac{65}{8}$

Method: $R = \frac{abc}{4A} = \frac{13 \cdot 14 \cdot 15}{4 \cdot 84} = \frac{2730}{336} = \frac{65}{8}$.

Investigate further: Forgetting to reduce, or using the semiperimeter instead of the area in the denominator.

Section C Substitution & Structure

1. **Answer:** C) $(1, 1)$

Method: $\left(\frac{1+3-1}{3}, \frac{-2+4+1}{3}\right) = (1, 1)$.

Investigate further: Slipping a sign in the y -average and getting $(2, 1)$ or $(1, 2)$.

2. **Answer:** A) 21

Method: $A = rs$ gives $s = A/r = 84/4 = 21$.

Investigate further: Confusing $A = rs$ as $A = s/r$ or dividing by twice r .

3. **Answer:** A) $\left(4, \frac{5}{3}\right)$

Method: The perpendicular bisector of $(0,0)$ – $(8,0)$ is $x = 4$. Let the centre be $(4, y)$: $(4)^2 + y^2 =$

$(4-4)^2 + (y-6)^2$ gives $12y = 20$, $y = \frac{5}{3}$.

Investigate further: Assuming the centre lies on an altitude and picking $(4, 3)$.

4. **Answer:** A) $\frac{\sqrt{5}}{2}$

Method: Euler's formula $d^2 = R^2 - 2Rr$ with $R = \frac{5}{2}$, $r = 1$: $d^2 = \frac{25}{4} - 5 = \frac{5}{4}$, $d = \frac{\sqrt{5}}{2}$.

Investigate further: Adding radii naively. The incenter and circumcentre coincide only in an equilateral triangle.

5. **Answer:** A) 6

Method: $BD : DC = 3 : 4$ with total 14: $BD = \frac{3}{7} \cdot 14 = 6$.

Investigate further: Picking the 4 part as if the ratio named the segments in the wrong order.

6. **Answer:** A) $\frac{2\sqrt{34}}{3}$

Method: $G = (\frac{0+8+2}{3}, \frac{0+0+6}{3}) = (\frac{10}{3}, 2)$; $OG = \sqrt{(\frac{10}{3})^2 + 2^2} = \frac{\sqrt{136}}{3} = \frac{2\sqrt{34}}{3}$.

Investigate further: Squaring the average instead of averaging the squares.

7. **Answer:** A) $(3, \frac{9}{4})$

Method: Altitude from C is $x = 3$. Side AC has slope $\frac{4}{3}$, so the altitude from B is $y = -\frac{3}{4}(x-6)$; at $x = 3$, $y = \frac{9}{4}$.

Investigate further: Returning the vertex $C = (3, 4)$ itself. The altitudes meet $\frac{9}{4}$ up, not 4.

8. **Answer:** A) $(4, \frac{8}{3})$

Method: On the Euler line $\overrightarrow{GH} = 2\overrightarrow{GO}$, so $H = G + 2(G - O)$; $G = (4, 2)$, $O = (4, \frac{5}{3})$ gives $H = (4, \frac{8}{3})$. Cross-check with the altitude method.

Investigate further: Using $\frac{1}{2}$ instead of 2 and landing on $(4, 2)$ mixed with O .

Section D Challenge Ramp

1. **Answer:** A) $\frac{21}{2}$

Method: Right triangle 9, 12, 15: $r = \frac{9+12-15}{2} = 3$, $R = \frac{15}{2}$, $r + R = \frac{21}{2}$.

Investigate further: Quoting the hypotenuse instead of R , or omitting the inradius.

2. **Answer:** A) 6

Method: $\frac{BD}{DC} = \frac{AB}{AC} = \frac{12}{16} = \frac{3}{4}$ and $BD + DC = 14$ give $BD = \frac{3}{7} \cdot 14 = 6$.

Investigate further: Breaking 14 in half for an isosceles reflex; the ratio offsets the midpoint.

3. **Answer:** A) $\frac{\sqrt{106}}{2}$

Method: Apollonius: $m_a^2 = \frac{2b^2+2c^2-a^2}{4}$ with $a = 8$, $b = 6$, $c = 7$: $m^2 = \frac{72+98-64}{4} = \frac{106}{4}$, $m = \frac{\sqrt{106}}{2}$.

Investigate further: Putting a, b, c into the wrong slots, or taking a half-scaled median.

4. **Answer:** A) $(1, 1)$

Method: Weight by opposite sides: $I = \frac{BC \cdot A + CA \cdot B + AB \cdot C}{a+b+c} = \frac{5(0,0) + 4(3,0) + 3(0,4)}{12} = \frac{(12,12)}{12} = (1, 1)$.

Investigate further: Weighting by the *adjacent* sides or cross-adding the coordinates.

5. **Answer:** A) $\frac{\sqrt{65}}{8}$

Method: $d^2 = R^2 - 2Rr = \frac{65^2}{64} - 2 \cdot \frac{65}{8} \cdot 4 = \frac{4225}{64} - \frac{4160}{64} = \frac{65}{64}$, so $d = \frac{\sqrt{65}}{8}$.

Investigate further: Dropping the $2Rr$ term, or reporting R as the answer.

Top 5 Patterns

- **The centroid is the vertex average.** Sum all three x -coordinates and all three y -coordinates, then divide by 3. (see A1, A6, B1, C1, C6)
- **Right triangles donate their centres for free.** The circumcentre is the hypotenuse midpoint, the orthocentre is the right-angle vertex, and $r = \frac{a+b-c}{2}$. (see A2, A3, A4, A7, B2, B3, B5, B9, D1)
- $A = rs$ and $R = \frac{abc}{4A}$ **bind sides to circles.** Every pair of sides and area plugs into these two formulas. (see A5, B8, B10, C2)
- **The centroid divides each median in the ratio 2 : 1.** The vertex-to-centroid part is twice the centroid-to-midpoint part. (see A10, B4, C8)
- **Angle bisector theorem: segments pair with the *adjacent* sides.** $\frac{BD}{DC} = \frac{AB}{AC}$. (see A8, B6, C5, D2)

Common Traps

- **The incenter is not at the average of the vertices.** The centroid equates masses; the incenter needs the *opposite-side* weighting. (see D4, C4)
- **A right circumcentre sits halfway along the hypotenuse, not at its own vertex-average.** Spot the right angle before averaging anything. (see A2, B5, D1)
- $A = rs$ and $R = \frac{abc}{4A}$ **ask for the same A .** Semi-perimeter, not perimeter, goes under A in $r = A/s$. (see B8, B10, C2)
- **Euler's formula is $d^2 = R^2 - 2Rr$, not $R - r$ or $R^2 - r^2$.** For a non-equilateral triangle the two centers are never glued together. (see C4, D5)

- **Apollonius and the bisector ratio are not interchangeable.** Medians bisect *lengths*, angle bisectors split *the opposite side in the side ratio*. (see D3, D2, A8, A9, B7)

“Every triangle centre is cooked by the same three sides: one averages them, one divides four times the area, one halves a hypotenuse.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths